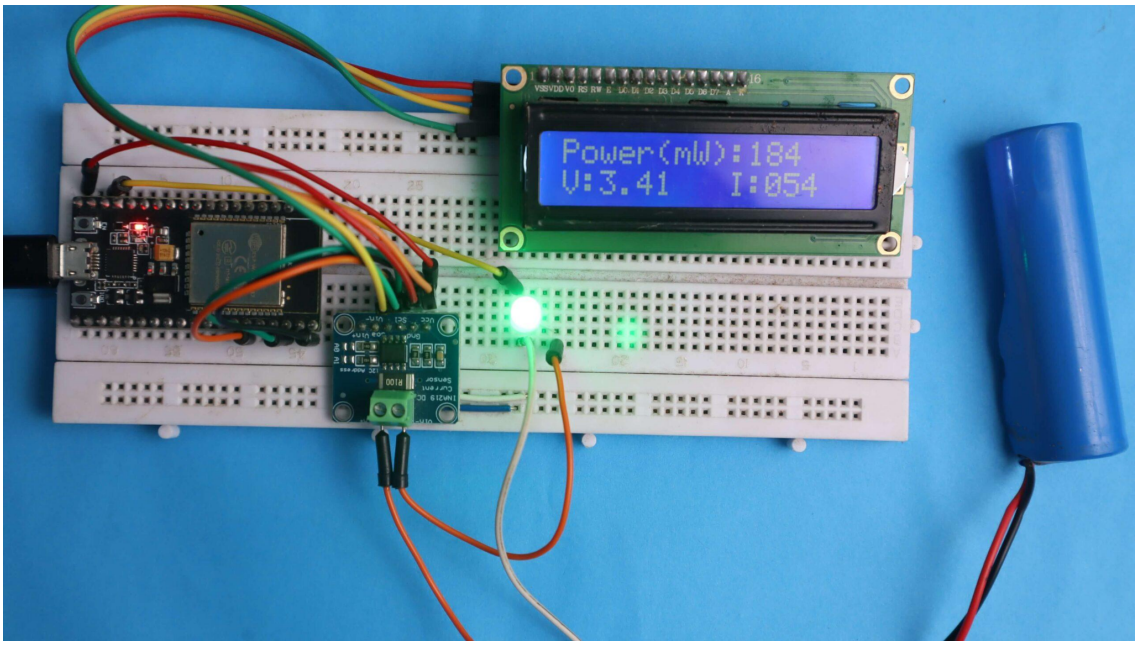
IoT Battery Health Monitoring Innovations

Concepts, emerging technologies, and a proposed final-year engineering project



Prototype image supplied by the user: ESP32-based battery measurement setup with an LCD, current-sensor module, LED, and battery.

Introduction

IoT-based battery health monitoring combines sensors, embedded controllers, cloud platforms, and artificial intelligence (AI) to monitor a battery's condition in real time. Unlike conventional systems that mainly measure voltage, modern solutions can estimate battery degradation, detect faults early, and help predict when maintenance or replacement may be necessary.

1. Key Innovations

1.1 Real-Time Battery Health Monitoring

Sensors measure battery voltage, current, and temperature. An ESP32 or similar microcontroller sends readings to a mobile or web dashboard, allowing users to monitor charging, discharging, and overheating remotely.

1.2 AI-Based State of Health (SoH) Prediction

Machine-learning algorithms analyse historical charging cycles, temperature, voltage, and current to estimate battery degradation and remaining useful life. This can help predict when a battery may need replacement.

1.3 Intelligent Thermal and Fault Detection

Temperature sensors and anomaly-detection algorithms identify overheating, abnormal voltage, excessive current, and possible cell imbalance. The system can send an alert and, with suitable protection hardware, disconnect charging when unsafe conditions occur.

1.4 Predictive Solar Charging

An IoT controller combines battery condition, solar generation, weather forecasts, and expected energy demand to optimise charging. It can prioritise charging before expected poor weather while avoiding unnecessary charging and excessive battery cycling.

1.5 Smart Cell Balancing and Degradation Tracking

For battery packs with accessible individual-cell measurements, the system detects voltage differences and supports cell balancing. Tracking these differences over time can reveal weak cells before they significantly reduce overall battery performance.

2. Emerging Innovations in Advanced Systems

Innovation	What makes it different
Remaining Useful Life (RUL) prediction	Estimates how much useful operating life a battery has left before reaching its end-of-life threshold.
Digital battery twins	Creates a software model of a physical battery to compare expected behaviour with real sensor readings.
Wireless battery monitoring	Uses wireless sensors to reduce wiring and simplify monitoring across larger battery installations.
Impedance-based health diagnosis	Analyses changes in internal electrical impedance to detect degradation that voltage readings alone may miss.
Cloud-based battery analytics	Stores historical data, compares battery performance over time, and supports remote maintenance decisions.

These approaches are relevant to modern battery management, particularly for solar energy storage and electric vehicles. Their usefulness depends on the battery chemistry, available sensors, quality of historical data, and validation of the estimation algorithms.

3. Suggested Final-Year Project

Project title: IoT-Based Smart Solar Battery Health Monitoring and Predictive Maintenance System

Core idea: Develop a system that monitors a solar battery in real time, estimates its health, detects abnormal conditions, and displays the information on a web dashboard.

Proposed Hardware

• ESP32 microcontroller with Wi-Fi

• Voltage and current sensors

• Battery temperature sensor

• Battery-compatible charging controller

• Cloud database and web dashboard

Proposed Software Features

• Real-time voltage, current, temperature, and state-of-charge monitoring

• Historical charging and discharging graphs

• Battery health estimation using capacity and operating history

• Overtemperature and abnormal-voltage alerts

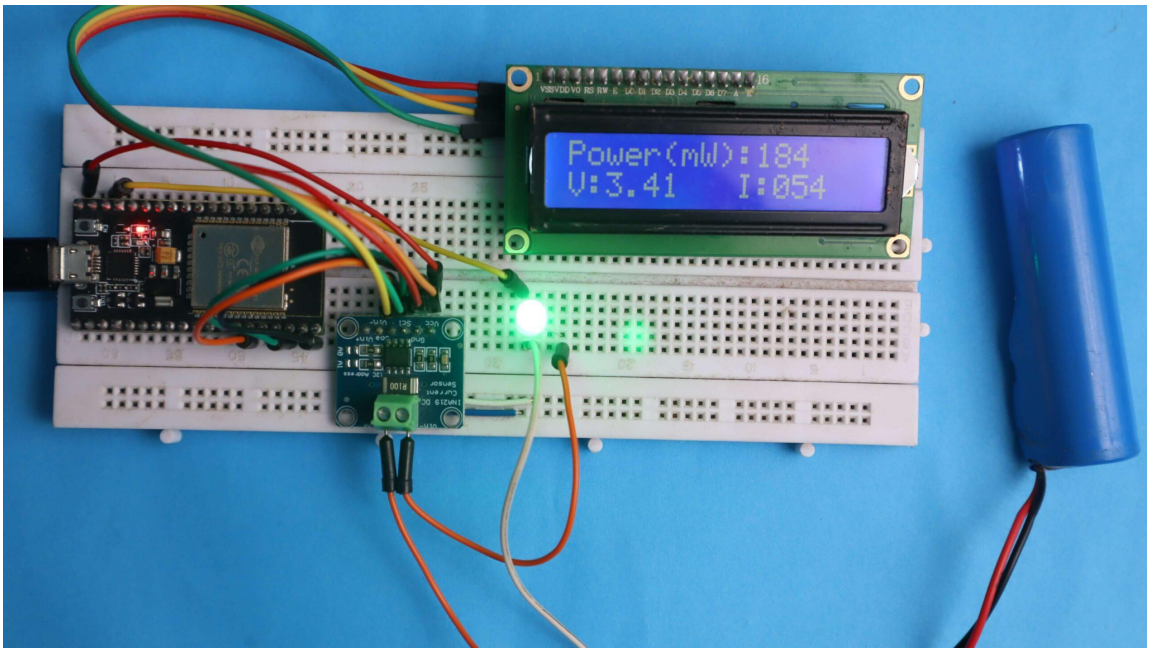
• Predictive maintenance using a machine-learning model

4. Recommended Project Focus

Make battery health prediction the main innovation, rather than simply displaying voltage and temperature. This creates a stronger engineering contribution by combining power electronics, embedded systems, IoT, and predictive analytics.

A basic ESP32 sensor dashboard is a monitoring system, not automatically an AI system. To demonstrate genuine prediction, collect battery-cycle data and validate the model against measured battery capacity or other reliable health indicators. Use independent battery protection circuitry rather than relying on software alerts alone.

5. Prototype Image and Relation to the Project



The supplied prototype shows an ESP32, an LCD displaying voltage/current/power readings, a current-sensor module, an LED, and a battery. It provides a useful starting point for the monitoring stage. The next steps are to add temperature sensing, log readings over time, and implement validated battery health estimation.

6. Suggested Development Sequence

Stage 1 — Sensor integration

Connect the ESP32, calibrate voltage and current measurements, and verify temperature readings.

Stage 2 — Monitoring and logging

Integrate the compatible charge controller and build a live dashboard with timestamped data.

Stage 3 — Health assessment

Implement state-of-charge estimation and conduct controlled capacity tests to calculate capacity-based state of health.

Stage 4 — Prediction and evaluation

Develop a prediction model, test fault alerts, compare estimates with measured results, and document limitations.

Safety note

Use a battery-compatible charge controller and protection circuit. Check sensor voltage/current ratings, polarity, and wiring before powering the setup. Do not rely on the ESP32 alone to prevent overcharge, over-discharge, or overheating.